Integration guide: Retro Horse Race (platform edition)

For the platform / operator that wants to offer the game to its players. It explains the integration as built today, how a session and a bet flow, what the platform must implement, and, at the end, **what is missing or weak and should be improved**. The wire format of the wallet calls is in `WALLET_API.md`; deployment is in `DEPLOY.md`.

1. Integration model

Item	Today
Launch	Redirect. The player's browser leaves the platform and opens `https://GAME_HOST/?token=<one-time token>`. No iframe / embed mode.
Wallet	Seamless (single) wallet. The platform keeps the money; the game debits each bet and credits each win through the platform's HTTP API.
Currency	One currency per game server (`CURRENCY`), amounts are integers in minor units (centavos).
Game rounds	One shared continuous table: bet (25 s) -> draw -> 6 s countdown -> race (30 s) -> results (10 s) -> next round. All players see the same race.
Result	Drawn by the game server (crypto RNG, committed with SHA-256 before betting closes, seed revealed after the race). Nobody, including the operator, can set the result in live mode.
Payout	Fixed odds board (shuffled each race). Payout = stake x odds, floor to a minor unit. Long-run RTP is an operator setting, 88% to 99%.

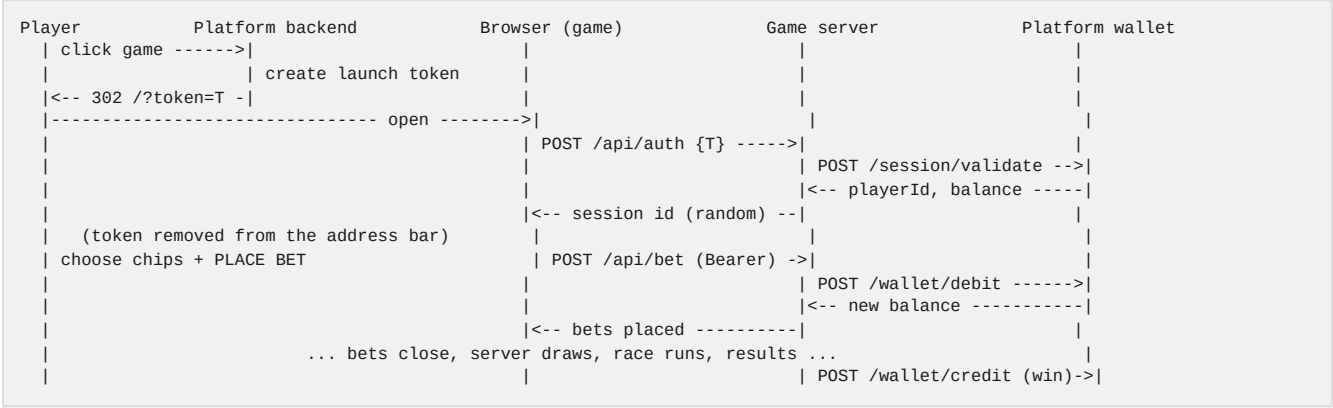
2. Roles

- **Platform backend**: creates launch tokens, implements the four wallet endpoints, keeps the player's balance.
- **Game server** (this repo): serves the game, runs rounds and draws, calls the wallet, keeps a ledger (`data/ledger.jsonl`).
- **Operator**: runs the game server and uses `/controller` (report and RTP setting).

3. Setup steps for the platform

1. **Deploy the game server** in live mode with the HTTP wallet (see `DEPLOY.md`): `GAME_MODE=live WALLET=http WALLET_URL=https://platform.example.com/horse-api WALLET_KEY=<secret> CONTROLLER_KEY=<operator key> CURRENCY=PHP RTP=90 node server.js` Put it behind HTTPS. Keep `/controller`, `/api/admin`, `/api/report`, `/api/command` reachable by the operator only.
2. **Implement the wallet API** (section 5).
3. **Add the launch button** in the lobby: your backend creates a random one-time token for the logged-in player and redirects the browser to `https://GAME_HOST/?token=<token>`.
4. **Test locally** without a platform: `GAME_MODE=live WALLET=mock node server.js`, open `http://localhost:8080/?token=demo-alice`.
5. **Reconcile** daily: compare `data/ledger.jsonl` (match on `txId`) with the platform's wallet transactions.

4. Flow of a session and a bet



Session details: the game server turns the launch token into its own random 192-bit session ID, valid for up to 12 hours (`sessionTtlMs`). It stores only a hash of the ID, in memory and in `data/sessions.json`, so a restart does not end sessions. The browser keeps the ID in `sessionStorage` and sends it as `Authorization: Bearer <session>`. Closing the tab ends the player's session on that device; the player launches again from the lobby. Balances and bets are not lost on a restart (money is on the platform; bets are recovered from the ledger and settled or refunded).

5. What the platform implements

All calls are `POST`, JSON, signed with HMAC (`X-Timestamp`, `X-Nonce`, `X-Signature`), amounts in minor units. Full detail in `WALLET_API.md`.

Endpoint	Request	Reply
`/session/validate`	`{token}`	`{ok, playerId, currency, balance, returnUrl}` plus optional `brandId`, `decimals`, `limits`, `realityCheckMinutes`, `blocked`; the token is single use
`/wallet/balance`	`{playerId}`	`{ok, balance}`
`/wallet/debit`	`{playerId, amount, txId, meta: {raceNo, kind:"bet"}}`	`{ok, balance}` or HTTP 402 `INSUFFICIENT_FUNDS`
`/wallet/credit`	`{playerId, amount, txId, meta: {raceNo, kind:"win"|"refund"}}`	`{ok, balance}`
`/wallet/rollback`	`{playerId, amount, txId, rollbackTxId, meta}`	`{ok, balance, status}`; cancels the debit `txId` and refuses it if it arrives later

Rules the wallet must follow:

- **Idempotent on `txId`**: the same `txId` twice is applied once and returns the same reply.
- A debit that times out may or may not have been applied. The game retries once with the same `txId`; if it still fails the bet is rejected, logged as `debit_unknown`, and rolled back through `/wallet/rollback` (retried until confirmed).
- Failed credits (wins, refunds) are retried forever with back-off.
- Serve the wallet API over HTTPS only, keep `WALLET_KEY` secret.

6. Settings the operator controls

Setting	Where	Notes
`RTP` (88 to 99%)	env at start, `/controller` while running	Takes effect from the next race. The odds never change; the three cheapest pairs absorb the difference.
`ODDS` / `ODDS_FILE`	env	Change the odds table (a restart is needed).
`BET_MS`, `MIN_BET`, `MAX_BET`, `MAX_ROUND_STAKE`	env	Minor units. Max bet is per pair, max round stake is per player per race.
`CURRENCY`, `WALLET_TIMEOUT_MS`	env	

7. Security as built

- Browser never sends balances or payouts; stakes are validated server-side (integer, min/max, 15 valid pairs, per-pair and per-round limits).
- Launch token is one time and removed from the URL; `Referrer-Policy: no-referrer` on pages.
- Results come from the server's RNG with a SHA-256 commit shown before betting closes and the seed revealed after the race.
- Wallet requests are HMAC-signed with timestamp and nonce; the mock wallet cannot run in production; unknown debits are rolled back; sessions are stored hashed and survive restarts.
- Players can be limited (bet limits, daily loss limit), blocked and shown reality checks; every bet is in a per-player history with a verifiable draw proof.
- Bet requests are rate limited (12 per 5 s per session); the controller cannot change results in live mode.

8. Improvements to consider (gaps today)

Ordered by importance for a real-money launch.

Must have before real money

1. **Signed wallet requests. DONE.** Every wallet call carries `X-Timestamp`, `X-Nonce` and an HMAC-SHA256 `X-Signature` (`server/sign.js`, spec in `WALLET_API.md`). The platform rejects bad signatures, stale timestamps and replayed nonces, and should also allowlist the game server's IP. `WALLET_AUTH=bearer` keeps the old mode for platforms that cannot verify yet.
2. **Block the mock wallet in production. DONE.** In live mode the play-money wallet needs `ALLOW_MOCK_WALLET=1` and is always refused when `NODE_ENV=production`; a `WALLET_KEY` shorter than 16 characters is refused too.
3. **Rollback of unknown debits. DONE.** When a debit times out twice the bet is rejected and the game calls `POST /wallet/rollback` for that `txId`, retrying with back-off (also after a restart) until the platform confirms. The platform must also refuse a late debit with a cancelled `txId`. If the platform has no rollback endpoint the case is left in the ledger for manual reconciliation. A status endpoint is still an option if you prefer it to rollback.
4. **Persistent sessions. DONE.** Sessions are kept in `data/sessions.json` (ids stored hashed, file mode 0600), so a restart or deploy does not log players out. Expired sessions are not restored. Operators can end a player's sessions with the controller command `closeSessions` (`POST /api/command {cmd:"closeSessions", playerId}`).

5. **Independent security review and RNG/math certification** by an accredited lab if the operator is licensed. (Not something the code can provide; still open.)

Should have

1. **Round and bet IDs in every wallet call. DONE.** Debit, credit and rollback carry `gameId`, `transactionType`, `roundId`, `betId` / `betIds`, `refTxId`, `currency` and `brandId` at the top of the body (and in `meta`).
2. **Player history and draw proof. DONE.** `GET /api/history` (player session), `GET /api/admin/history?playerId=` (operator) and `GET /api/round/<n>` (public proof: commit, seed, odds, result). The betting panel has a MY HISTORY list. History is rebuilt from the ledger after a restart.
3. **Operator-driven limits. DONE.** `/session/validate` may return `limits` (min / max bet, max per race, daily loss limit) and `blocked`. A daily loss limit blocks bets that would take the player's net loss of the UTC day over it. Operator commands `closeSessions`, `blockPlayer` and `unblockPlayer` end sessions and keep a player out (self-exclusion).
4. **Reality-check hook. DONE.** `realityCheckMinutes` (per player or `REALITY_CHECK_MIN`) shows time played and the session result every N minutes, with CONTINUE and LEAVE (to `returnUrl`). The BACK TO LOBBY link is always there.
5. **Server-to-server events. DONE.** Signed `round.settled` and `big.win` webhooks (`WEBHOOK_URL`), at least once, stable `id`, retried with back-off and recovered after a restart.
6. **Multi-currency and multi-brand. DONE.** Currency, decimals and brand come per session; per-currency limit tables (`CURRENCY_LIMITS`); the report shows staked and paid per currency. **Multi-client (several platforms on one game server). DONE (registry and client API).** Each client has its own wallet URL and signing secret, webhook, API keys (hashed, rotating, optional IP allow list), currency, limits and return-link hosts; players are namespaced per client; a client API (`/api/client/...`) gives each client its own report, history and player controls. See `CLIENT_API.md`. All clients share one race and one RTP. A back-office web panel for the clients (`/backoffice`: own logins, roles viewer / support / admin, optional 2-step sign-in, reports, players, player controls, audit log) is built: see `BACKOFFICE.md`. The operator manages clients, keys, secrets, limits, webhooks and client users from an admin console (`/admin`, personal accounts with roles and 2-step sign-in): see `ADMIN_CONSOLE.md`.
7. **Bonus / free-bet support.** Today there is only real-money debit and credit.

Nice to have

1. **Embed mode.** An iframe launch with a `postMessage` API (balance updated, round result, close game). The game already emits `countdown`, `start` and `result` events to a parent window, but the betting panel and session flow are built for redirect.
2. **Tamper-evident ledger. DONE.** Every ledger line is hash-chained (`node scripts/verify-ledger.js`); a broken chain is detected at start-up, raises an alert and turns `/healthz` red. Each `round.settled` webhook carries the newest hash (`ledgerHash`), so the platform can anchor the ledger outside the game server. Shipping the file to an append-only store is still up to the operator.
3. **Monitoring. DONE.** `/metrics` (Prometheus), `/healthz` that reflects the real state, a Health and alerts panel in the controller, and built-in alerts to a webhook: stuck wins / refunds / rollbacks, failing or slow wallet, stalled round loop, broken ledger, payouts drifting from what the odds predict, low disk.
4. **High availability. DONE (active / standby).** A lease file keeps exactly one instance running the races; a standby takes over in about 10 to 15 seconds from the same data directory, replaying the ledger; graceful shutdown on deploys. This is failover, not load sharing: still one table, one process.
5. **Aggregator adapter.** A small adapter in `server/wallet.js` if the platform sits behind a casino aggregator with its own API format.
6. **Localisation and currency formatting** per player, and a launch parameter for language.

9. Questions to settle with the platform

1. Redirect only, or do you also need an iframe embed?
2. Which wallet API do you already have? (We can adapt `server/wallet.js` to it.)
3. Do you need bonus, multi-currency or per-player limits at launch?
4. What must the player history show, and how do you want to reconcile (daily file, API)?
5. Who hosts the game server, and who operates `/controller`?
6. Is the operator licensed, and does that require lab certification of the RNG and the game math?